Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Advanced Pre-Chamber Ignition Systems for Hydrogen-Fueled Internal Combustion Engines			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Energy and Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile/Rail Vehicles	Engines and Power Transmission Systems	New and Renewable Energy	Hydrogen
Concept of Proposed Technology	The proposed technology focuses on the development and experimental validation of advanced pre-chamber ignition systems for high-efficiency internal combustion engines, with a primary emphasis on hydrogen operation. The work addresses the systematic design, evaluation, and thermal assessment of pre-chamber geometries and material concepts using a combination of experimental testing and numerical simulation methods. A key characteristic of the proposed approach is the high degree of design flexibility in pre-chamber geometry. Such flexibility can be realized through different manufacturing routes, including additive and conventional manufacturing processes. This technology openness allows the ignition concept to be adapted to various industrial capabilities and cost constraints while still enabling optimized combustion performance and targeted thermal management. The pre-chamber ignition system is intended to be applicable to a wide range of fuels. While the primary focus is on hydrogen, the developed concepts and methods are expected to be transferable to other alternative and low-carbon fuels. This addresses current and future requirements for efficient and low-emission combustion systems.			

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	Experimental activities include the evaluation of instrumented passive pre-chamber spark plugs on an engine test bench using passenger car and motorcycle engines as representative platforms. The system will be assessed with respect to combustion efficiency, engine performance, emission behavior, and thermal robustness under representative part-load and full-load operating conditions. Spatially resolved temperature measurements using integrated thermocouples enable a detailed characterization of the thermal behavior during engine operation. In addition to experimental investigations, finite element (FE) simulations will be performed to analyze heat transfer mechanisms and thermal load distributions within the pre-chamber ignition system. Numerical results will be correlated with experimental data to improve predictive capability and to support geometry and material optimization. Data-driven evaluation methods will further enhance the understanding of the relationship between operating conditions, thermal loads, and component durability. Potential applications include hydrogen-fueled combustion engines in automotive and light-duty applications, where high efficiency, low emissions, and robust thermal management are critical. While larger industrial engines are not directly investigated within this project, the derived methodologies and design guidelines are expected to be transferable.
Rationale for Joint R&D Project	The development of advanced pre-chamber ignition systems for hydrogen combustion engines requires the close integration of design, manufacturing, and experimental validation competencies. Hydrogen operation is associated with fast combustion processes, high thermal loads, and strict emission requirements, which impose demanding boundary conditions on ignition components with respect to geometry, materials, and manufacturability. A joint R&D approach is essential because these challenges cannot be addressed by a single organization alone. The research institute contributes extensive expertise in engine testing, combustion analysis, thermal instrumentation, and numerical modeling, enabling a detailed experimental and analytical evaluation under realistic operating conditions. However, the institute does not maintain in-house manufacturing capabilities for advanced pre-chamber components. The Korean SME partner complements these capabilities by providing specialized know-how in the manufacturing and material implementation of pre-chamber ignition components. This includes the realization of complex geometries and advanced material concepts using suitable manufacturing technologies, as well as the ability to iteratively adapt designs based on experimental feedback. Through close collaboration, manufacturing-related design decisions can be directly informed by experimentally validated performance and thermal data. This significantly reduces development risks, accelerates technology maturation, and ensures that the resulting pre-chamber ignition systems are both functionally validated for hydrogen combustion and suitable for industrial application.
Key Words	1. Pre-chamber ignition 2. Hydrogen combustion 3. Thermal management 4. Advanced ignition systems

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	The objective of this joint R&D project is to experimentally demonstrate and validate a passive pre-chamber ignition system on an engine test bench using passenger car and motorcycle engines, supported by numerical thermal analysis, and to derive transferable design and operation guidelines for hydrogen combustion applications. Specific objectives are: • Demonstration of stable and repeatable pre-chamber ignition under selected part-load and full-load operating conditions. • Quantitative assessment of the impact of pre-chamber ignition on engine efficiency, performance, and emission behavior relative to a conventional spark plug baseline. • Experimental determination of characteristic temperature levels, gradients, and thermal load profiles of the pre-chamber ignition system. • Validation of finite element–based thermal models using experimentally measured temperature data. • Derivation of transferable design and optimization guidelines applicable across fuels and manufacturing routes.			
R&D Target Specifications	The project aims to achieve the following target specifications: • Quantified changes in engine efficiency and performance relative to a baseline ignition system at defined operating points. • Comparative evaluation of relevant exhaust gas emissions (e.g., NOx, unburned hydrogen) under identical operating conditions. • Identification of critical thermal load cases based on spatially resolved temperature measurements. • Demonstrated agreement between numerical thermal models and experimental temperature trends. • Documented design criteria for pre-chamber geometry and material selection.			
R&D Plan	The project is structured into five sequential phases, progressing from laboratory-scale validation to system-level validation on passenger car and motorcycle engine test benches. Phase 1: Engine adaptation, baseline commissioning, first pre-chamber design (approx. 6 month) Adaptation of the base engine to accommodate the pre-chamber concept, assembly and commissioning of the engine test bench, and establishment of baseline operation using conventional spark plugs. In parallel: concept and design of the initial pre-chamber sparkplugs Phase 2: Manufacturing of the initial pre-chamber spark plugs (approx. 6 month). Development of the manufacturing process for the hydrogen-specific design. Decisions regarding the use of additive or conventional manufacturing methods, the base spark plug and the machining and welding methods, as well as the manufacturing of the first baseline prototypes. In parallel, baseline engine mapping measurements will be performed with conventional spark plugs. Phase 3: Integration of pre-chamber ignition system and instrumentation (approx. 3 months) Integration of manufactured pre-chamber spark plugs, installation of thermal and performance-related instrumentation, and commissioning of the instrumented engine. Phase 4: Experimental Investigation – Baseline Pre-Chamber Design (approx. 9 months) - Engine mapping at selected part-load and full-load operating points - Analysis of performance, emissions and thermal behavior - Development and correlation of FE-based thermal models - Mechanical post-test inspection of components The design will be optimized based on the experimental results and the experiences of the first manufacturing process.			

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	Phase 5: Experimental investigation – optimized pre-chamber design (approx. 12 months) Evaluation of an optimized pre-chamber design following the same methodology as Phase 4, enabling direct comparison and refinement of design guidelines and numerical models.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the conceptual development, experimental validation, and thermal assessment of the pre-chamber ignition system. Its main responsibilities include: • Definition and evaluation of pre-chamber ignition concepts, including geometry and material-related design aspects • Adaptation and operation of the engine test bench, including engine modification, commissioning, and stable operation under defined test conditions • Integration and operation of instrumented pre-chamber spark plugs, including thermal and performance-related measurement systems • Experimental investigation of engine performance, combustion efficiency, and emission behavior at selected operating points • Thermal characterization of the pre-chamber ignition system based on spatially resolved temperature measurements • Development and validation of finite element–based thermal models and data-driven evaluation methods • Analysis, interpretation, and documentation of experimental and numerical results • Derivation of design, evaluation, and optimization guidelines for pre-chamber ignition systems The research institute will provide the required infrastructure, including the engine test bench, measurement equipment, data acquisition systems, and qualified scientific personnel.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the manufacturing-related development and provision of the pre-chamber ignition components. Its main responsibilities include: • Development and manufacturing of pre-chamber spark plug prototypes based on jointly defined design specifications • Selection and implementation of suitable manufacturing routes, including additive or conventional manufacturing technologies • Integration of advanced material concepts and multi-material solutions, where applicable • Iterative refinement of pre-chamber geometry and material configuration based on experimental feedback • Provision of technical documentation related to manufacturing processes and material properties • Close collaboration with the research institute throughout the project, including joint evaluation of results and alignment on optimization strategies

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver experimentally validated pre-chamber ignition concepts enabling stable, efficient, and low-emission hydrogen combustion. Passenger car and motorcycle engines serve as representative and cost-efficient validation platforms for hydrogen pre-chamber ignition concepts at this development stage. The results will support the development of hydrogen engines with improved thermal robustness and provide Korean SMEs with validated design insights transferable to industrial product development.

6. References

(1) Trapp C., Joshi V., Alkezbari A., Rottenkolber G., Kottakalam S., Nenzel M. (2025) Methodology for the Development of Passive Prechamber Spark Plugs for Use in Ultra-Lean Light-Duty Hydrogen Engines, 20th Symposium „Sustainable Mobility, Transport and Power Generation", Graz, Austria

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☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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